Secure Campus Network Using Cisco Packet Tracer

Mini Project Report and Implementation Evidence

1. Project Objectives

The primary objective of this project is to design and implement a secure campus network using Cisco Packet Tracer, ensuring reliable communication between two LAN networks while protecting network devices from unauthorized access.

- To design and simulate a campus network using Cisco routers, switches, and PCs.
- To configure IP addressing and enable communication between two LAN networks.
- To secure router and switch access using console and enable passwords.
- To implement SSH for secure remote device administration.
- To enable password encryption for improved security.
- To configure port security using sticky MAC address learning to prevent unauthorized device connections.
- 7. To disable unused ports and verify network connectivity and security configurations.

2. Network Design and Devices

The topology uses one Cisco 2911 router (R1), two Cisco 2960 switches (S1 and S2), and six PCs. R1 connects two IPv4 networks: 192.168.10.0/24 and 192.168.20.0/24. S1 serves PC1–PC3, while S2 serves PC4–PC6. Copper straight-through cables are used for the links.

3. Implementation Steps

Step 1 — Build the topology: Place R1, S1, S2 and six PCs in the Packet Tracer workspace. Connect R1 G0/0 to S1 G0/1 and R1 G0/1 to S2 G0/1. Connect each switch's Fa0/1–Fa0/3 ports to its three PCs.

Step 2 — Configure the router: Set hostname R1, assign 192.168.10.1/24 to G0/0 and 192.168.20.1/24 to G0/1, and enable both interfaces with no shutdown. Use show ip interface brief to inspect interface status.

Step 3 — Secure router access: Configure a console password under line console 0 and use enable secret for privileged EXEC access. Enable service password-encryption to obfuscate applicable line passwords in the configuration.

Step 4 — Enable SSH on R1: Configure the domain name campus.local, create a local admin account with a secret, generate RSA keys, select SSH version 2, and set the VTY lines to login local and transport input ssh.

Step 5 — Configure S1 and S2: Set hostnames, enable secrets, password encryption, management IP addresses on VLAN 1, and the corresponding default gateways. Configure console access and SSH on both switches using local credentials and RSA keys.

Step 6 — Enable port security: On PC-facing switch ports Fa0/1–Fa0/3, set access mode, enable port security, allow one secure MAC address, enable sticky MAC learning, and set violation mode to shutdown. Apply the equivalent settings on S2.

Step 7 — Configure PC addresses: PC1 192.168.10.10, PC2 .11, PC3 .12 with gateway 192.168.10.1; PC4 192.168.20.10, PC5 .11, PC6 .12 with gateway 192.168.20.1. Use subnet mask 255.255.255.0 for all PCs.

Step 8 — Verify and save: Test pings between PCs and router interfaces; test SSH access to the router and switches; inspect port security with show port-security and show port-security address; shut down unused ports Fa0/4–Fa0/24; save configurations using copy running-config startup-config.

4. IP Addressing Plan

R1 G0/0 — 192.168.10.1/24;
R1 G0/1 — 192.168.20.1/24.
S1 VLAN 1 — 192.168.10.2/24, gateway 192.168.10.1.
S2 VLAN 1 — 192.168.20.2/24, gateway 192.168.20.1.
PC1–PC3 use 192.168.10.10–12/24 and gateway 192.168.10.1.
PC4–PC6 use 192.168.20.10–12/24 and gateway 192.168.20.1.

5. Evidence and Observations

The supplied screenshots show the Packet Tracer topology, router interface configuration/status output, and switch CLI output associated with port-security configuration and a port-security violation/recovery sequence. The topology screenshot shows the router connected to two switches and six PCs. The router CLI screenshots show interface status/configuration output. The switch CLI screenshot includes a port-security violation message. These screenshots document implementation activity; they do not by themselves prove every listed ping and SSH test succeeded.

6. Security Features Used

Console password: protects local console login.

Enable secret: protects privileged EXEC mode.

service password-encryption: obfusc

ates applicable plaintext line passwords in the configuration, but is not strong cryptographic protection.

SSH version 2: provides encrypted remote administration.

Port security with sticky MAC learning: restricts each PC-facing port to one learned/allowed MAC address.

Violation mode shutdown: places the port into an error-disabled/shutdown state when the security violation occurs. Shutting unused ports reduces opportunities for unauthorized connections.

7. Testing Checklist

Record the actual result after running each test in Packet Tracer:

• PC1 → 192.168.10.1 ping

• PC1 → PC4 (192.168.20.10) ping

• SSH from PC1 to S1 (192.168.10.2)

• SSH from PC1 to R1 (192.168.10.1)

• SSH from PC4 to S2 (192.168.20.2)

• show port-security / show port-security address

• Demonstrate an unauthorized MAC-address violation

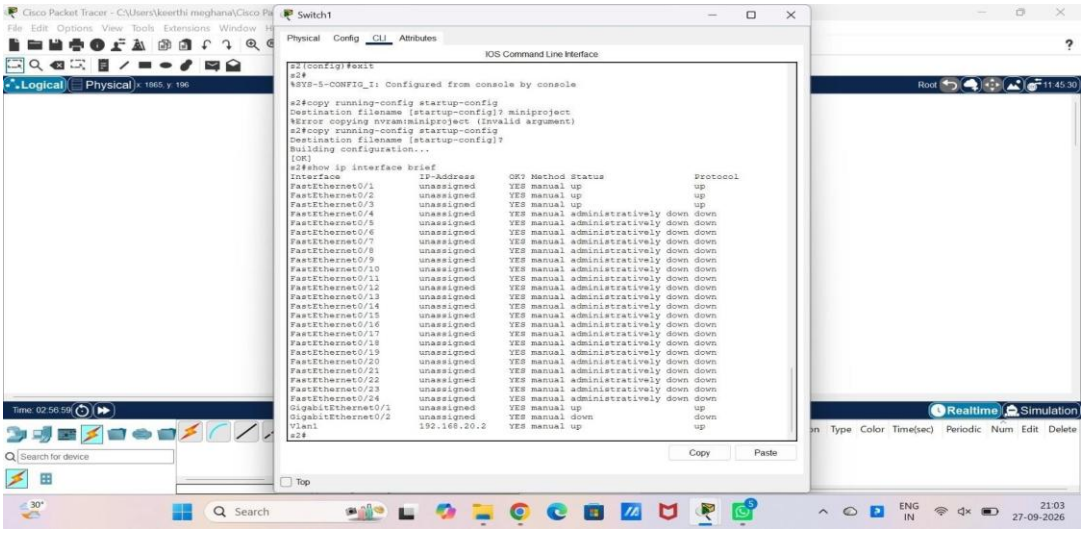
• Confirm unused switch ports are shutdown

Only mark a test as successful after observing the corresponding output.

8. Screenshots and Steps Evidenced

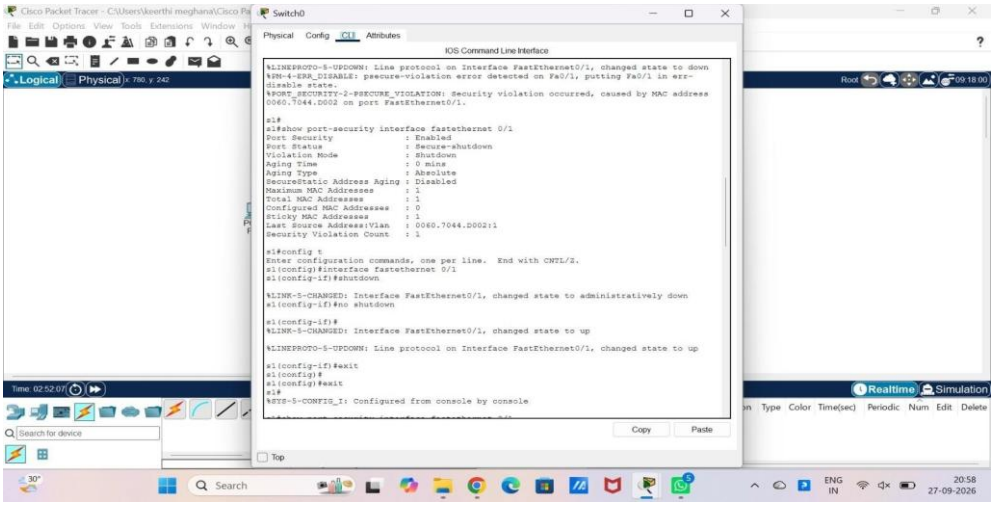
Screenshot 1 — Switch port status

The switch CLI displays interface status and port information. Use this image as evidence of inspecting switch interfaces.



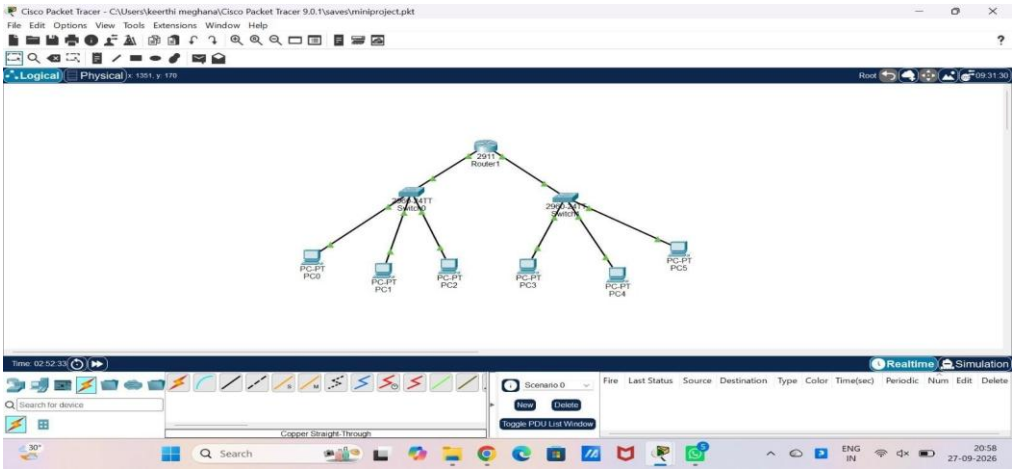
Screenshot 2 — Port-security violation

The CLI output includes a port-security violation message and follow-up interface commands. This supports the port-security demonstration.



Screenshot 3 — Network topology

The Packet Tracer workspace shows one router, two switches and six PCs connected in the intended two-LAN layout.



Screenshot 4 — Router interfaces

The router CLI shows interface configuration and the show ip interface brief command/output.

```
R1(config)#ip domain-name campus.local
R1(config)#username admin
R1(config)#username admin p
R1(config)#username admin pri
R1(config)#username admin privilege 15 se
R1(config)#username admin privilege 15 secret Admin@123
R1(config)#crypto k
R1(config)#crypto key gene
R1(config)#crypto key generate rsa
The name for the keys will be: R1.campus.local
Choose the size of the key modulus in the range of 360 to 4096 for your
  General Purpose Keys. Choosing a key modulus greater than 512 may take
  a few minutes.

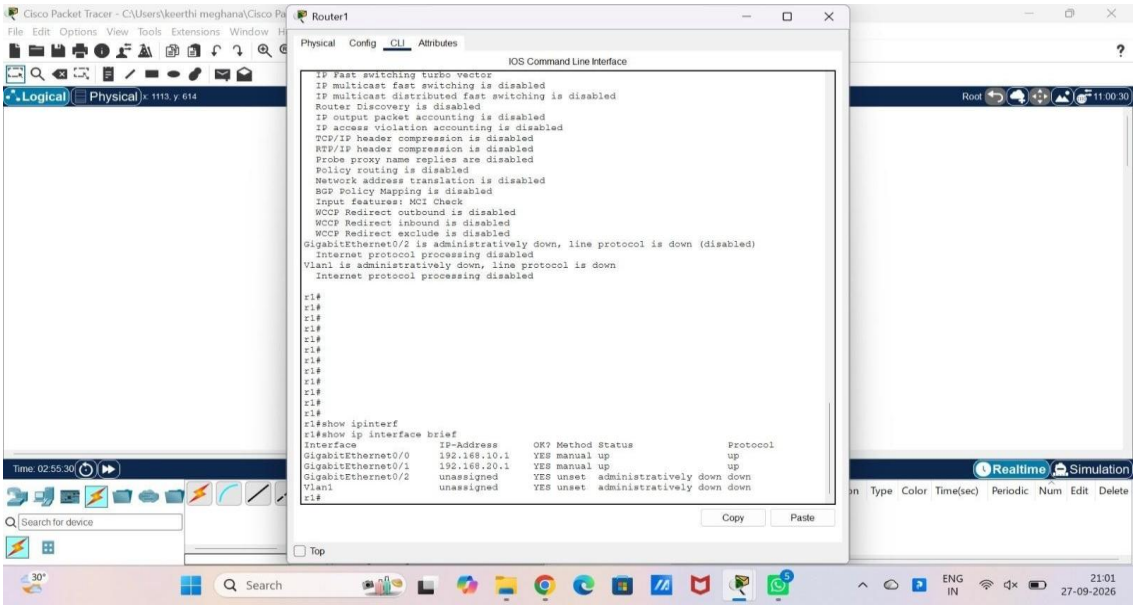
How many bits in the modulus [512]: 1024
% Generating 1024 bit RSA keys, keys will be non-exportable...[OK]

R1(config)#ip ssh
*Mar 1 0:14:9.172: %SSH-5-ENABLED: SSH 1.99 has been enabled
R1(config)#ip ssh version 2
R1(config)#line vty 0 4
R1(config-line)#login lo
R1(config-line)#login local
R1(config-line)#trans
R1(config-line)#transport inpu
R1(config-line)#transport input ssh
R1(config-line)#exit
R1(config)#exit
R1#
%SYS-5-CONFIG_I: Configured from console by console

R1#show ip ssh
SSH Enabled - version 2.0
Authentication timeout: 120 secs; Authentication retries: 3
```

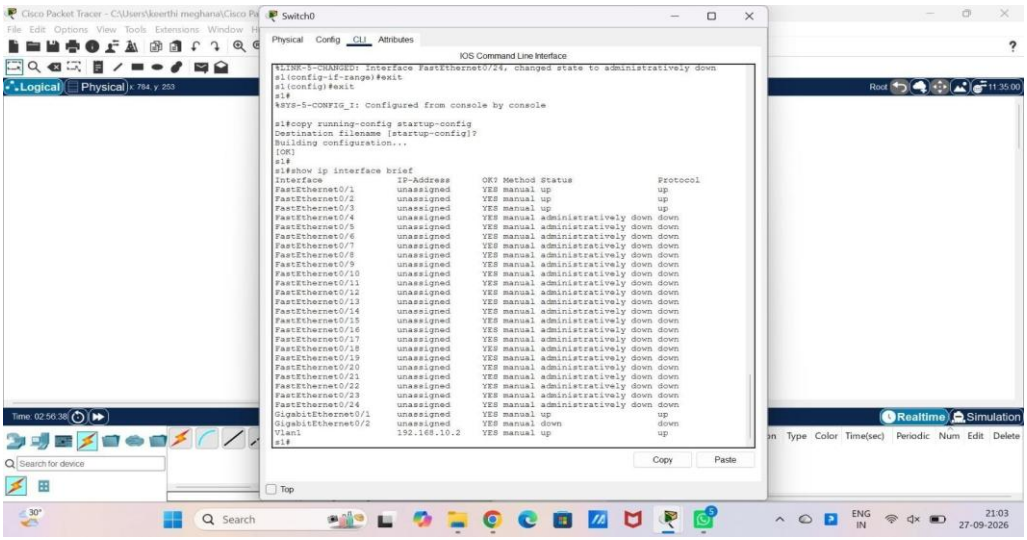
Screenshot 5 — Router status

The router CLI displays interface/status information used during configuration verification.



Screenshot 6 — Switch status

The switch CLI displays interface/status output for inspection.



9. Conclusion

The Secure Campus Network Using Cisco Packet Tracer project successfully demonstrates the design and implementation of a basic secured network environment for a university department. By configuring one router, two switches, and six PCs, the project establishes communication between two different LAN networks while incorporating essential network security measures.

Through connectivity testing, SSH verification, and port security violation demonstrations, the project provides practical exposure to Cisco networking concepts and security configurations. Overall, this project demonstrates how multiple basic security mechanisms can work together to improve network reliability, access control, and protection in a small campus network environment. It also builds a strong foundation for understanding and implementing network security in larger organizational networks.